

GROUP-5 LAB SECTION-4 MILESTONE-1

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Problem Statement:

Resource extraction operations in hazardous environments need a way to autonomously navigate unknown terrain to identify and retrieve high-value objects (erasers), especially where long distance communication is unreliable. Thus, the objective of this project is to design and build an autonomous mobile robot capable of collecting pencil erasers from a defined area, separating them by colour, and returning them to a base location within a fixed time limit.

Constraints & Ways to Measure:

ID	Constraints	How to Measure
1.	Size	The robot must be able to fit in the MSE lab locker.
2.	Sorting of Erasers	The picked-up erasures must be sorted either on the robot or on a stationary sorting system, and must be done based on wavelength (colour)
3.	Navigation	The robot must be able to traverse the area of operations without any hindrance or help from external sources.
4.	Power/Battery	The robot must be able to perform its entire task on a single charge of the batteries used, and the batteries used must not be modified.
5.	Weight Drivable	The robot must be able to move on its own weight and be light enough to be carried by a human (< 3 kg).
6.	Time-Efficient	Perform the entire operation within the limit of 2 minutes
7.	Autonomous	Operations must be performed without any human intervention or wireless communication.
8.	Budget	The overall cost of components for the robot, excluding components provided by the MSE 2202 lab kit, must not exceed \$50. The overall cost for FDM and laser cutting must not exceed \$20.
9.	Durability	The robot must be able to perform its task without the expense of its components. The robot must be able to perform the entire operation > 50 times without any need for component replacement.
10.	Manufacturability	The robot must be manufactured using available materials/components. This includes 4x MSE 2202 lab kit, FDM printed ABS/PETG/PLA components, and external components adhering to the budget constraint.

- Our design was developed directly around the defined system constraints to ensure measurable feasibility and performance. Size, weight, and manufacturability limits required the robot to fit within the MSE lab locker, remain under 3 kg, and be built using approved lab materials and FDM printing. Power and autonomy constraints required the system to complete the entire task on a single battery charge without human intervention or wireless assistance. Budget and durability limits further ensured the solution remained cost-effective while reliably performing at least 50 full operation cycles within the two-minute time requirement.

Objectives & Ways to measure:

ID	Objectives	How to Measure
1.	Simplicity of Design	The robot should be made with a minimum no. of components to increase reliability and durability
2.	Sorting Accuracy & Amount	Sort accuracy (the sorting between high-value and low-value erasers) should be > 85%.
3.	Efficient Weight Distribution	The robot should be balanced and must not stumble under its own weight.
4.	Navigation Accuracy	Should traverse the operation without losing sight of goals and have at least 90% success rate on test runs.
5.	Efficient UI	Clear indication of each process step and a failsafe in case of incidents or wrong performance of the task
6	Budget	The total cost of external components should be < \$20 provided by the project.

- Our design objectives focused on achieving high performance while maintaining simplicity and cost control. The system was engineered with a minimal number of components to improve reliability, while targeting over 85% sorting accuracy and at least 90% navigation success rate across multiple trials. Careful weight distribution ensured stable movement without imbalance or stalling during operation. Additionally, an efficient and clear user interface was implemented with process indication and fail-safe mechanisms, all while maintaining external component costs below the \$20 budget constraint.

Rough Concept Generation:

ID	Concept	Readiness (Keep/Drop)	Reasoning
Robot Movement			
1.	Gears with Tracks	Keep	
2.	Aftermarket Rubber Wheels	Keep	
3.	Self-Manufactured 3D Printed Wheels + Rubber Band Tires	Keep	
Chassis Design			
1.	4 Wheel Drive	Keep	
2.	2 Wheel Drive	Keep	
	S Wheel Drive + Dedicated Steering	Drop	Too complex with limited benefits.
Eraser Sorting			
1.	Conveyor Belt + 2 Lane Knocking System	Keep	
2.	Steep Slide with Hatch System	Keep	
3.	Slide With Sorting Wheel	keep	
4.	Angle Slide with Sensor Knocking System	Drop	It requires very precise timing and multisystem cooperation.
Eraser Drop Mechanism			
1.	Slide	Keep	
2.	Conveyor Belt	Drop	Require too much power and components
3.	Gate	Keep	
Eraser Pick Up Sensing			
1.	Camera	Drop	ESP32 does not have enough computational resources for image processing.
2.	UltraSonic Sensor	Keep	
3.	Infrared Sensor	Keep	

4.	UltraSonic + Infrared Sensor	Keep	
Eraser Pick Up Mechanism			
1.	Scoop pick-up	keep	
2.	Sticky Conveyor Belt	Drop	The Stickiness might lose its effectiveness over time
3.	Claw	keep	
4.	Vacuum Suction	Drop	Require many additional components on the robot
5.	Stabbing needle arm	Drop	Requires too much precision, too slow
6.	Scoop pick-up + Gate	Keep	

- During rough concept generation, multiple movement, sorting, sensing, and pick-up mechanisms were evaluated based on complexity, power consumption, reliability, and integration feasibility. Concepts were kept if they provided clear functional benefits while remaining simple to implement within hardware and software limitations. Designs were dropped when they introduced excessive mechanical complexity, required high computational resources, or demanded precise multi-system coordination with limited performance improvement. Overall, priority was given to energy-efficient, lightweight, and modular solutions that align with system constraints and practical implementation within the project scope.

Go/No-Go Matrix:

ID	Concepts	Size	Sorting of Erasers	Navigation	Power/Battery	Weight Drivable	Time-Efficient	Autonomous	Budget	Durability	Manufacturability	Reasoning
1	Gears with Tracks	Go	N/A	Go	N/A	Go	N/A	N/A	Go	Go	Maybe	- Requires precise alignment, tension control, and additional mechanical components, increasing fabrication complexity. - Higher material cost reduces overall efficiency and mobility compared to simpler wheel-based systems.
2	Aftermarket Rubber Wheels	Go	N/A	Go	N/A	Go	Go	N/A	Maybe	Go	Go	- Purchasing pre-manufactured components increases total project cost. - Cost may not justify marginal traction improvement over lower-cost alternatives.
3	Self-Manufactured 3D Printed Wheels + Rubber Band Ties	Go	N/A	Go	N/A	Go	N/A	N/A	Go	Maybe	Go	- 3D printed materials may wear down under continuous friction and load. - Rubber bands can stretch or degrade, reducing long-term reliability.
4	4 Wheel Drive	Go	N/A	Go	No Go	Go	Go	No Go	Go	Go	Go	- Requires synchronization between two ESP32 boards, increasing software complexity. - Coordinating encoder feedback and motor control adds unnecessary programming overhead.
5	2 Wheel Drive	Go	N/A	Go	Go	Maybe	Maybe	Go	Go	Go	Go	- Uneven weight distribution may reduce traction and stability. - Limited drive contact points can affect load handling performance.
6	Conveyor Belt + 2 Lane Knocking System	No Go	Go	Go	Go	Go	Maybe	Go	Maybe	Maybe	Maybe	- System footprint exceeds locker size constraints. - Additional belt and lane structure increases spatial requirements.
7	Steep Slide with Hatch System	Go	Go	N/A	Go	Go	Go	Go	Go	Go	Go	
8	Slide With Sorting Wheel	Go	Go	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Sorting wheel timing must align precisely with item arrival. - Rotational misalignment may reduce throughput efficiency.
9	Slide	Go	N/A	N/A	Go	Go	Go	Go	Go	Go	Go	
10	Gate	Go	N/A	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Servo actuation introduces delay during opening and closing cycles. - Mechanical lag may slow repeated sorting operations.
11	UltraSonic Sensor	Go	Go	Go	Go	Go	Go	Go	Go	Go	N/A	
12	Infrared Sensor	Go	Go	No Go	Go	Go	Go	Go	Maybe	Go	N/A	- Lab IR sensor is unreliable for accurate distance measurement. - Readings are affected by ambient light and surface reflectivity.
13	UltraSonic + Infrared Sensor	Go	Go	Go	Go	Go	Go	Go	Go	Go	N/A	
14	Scoop pick-up	Go	Go	N/A	Go	Go	Go	Go	Go	Go	Go	
15	Claw	Go	Go	N/A	Go	Maybe	Maybe	Go	Go	Maybe	Maybe	- Requires precise servo calibration and grip control. - Complex geometry increases fabrication difficulty.
16	Scoop pick-up + Gate	Go	Go	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Combining scoop and gate may introduce redundant handling steps. - Additional mechanism coordination increases cycle time.

- The final concept screening matrix evaluated each subsystem against key constraints, including size, power consumption, autonomy, budget, durability, and manufacturability. Concepts were marked “Go” when they met performance requirements without introducing excessive mechanical or software complexity. At the same time “No-Go” options were eliminated due to high power demand, spatial constraints, or synchronization challenges. “Maybe” selections reflect feasible solutions that carry moderate risks in durability, timing precision, or fabrication complexity. Overall, the matrix guided the selection of energy-efficient, lightweight, and practically implementable subsystems aligned with project constraints and performance objectives.

Pugh Matrix for Normalizing Weights:

Objectives	Simplicity of Design	Sorting Accuracy & Amount	Efficient Weight Distribution	Navigation Accuracy	Efficient UI	Budget	Total	Normalized Weighting	Legend
Simplicity of Design	1.00	0.14	1.00	0.20	9.00	0.14	11.49	0.12	Relative Importance
Sorting Accuracy & Amount	7.00	1.00	7.00	1.00	9.00	1.00	26.00	0.27	Exceptionally Less
Efficient Weight Distribution	1.00	0.14	1.00	0.20	9.00	0.33	11.68	0.12	Much Less
Navigation Accuracy	5.00	1.00	5.00	1.00	9.00	0.20	21.20	0.22	Less
Efficient UI	1/9	1/9	1/9	1/9	1.00	0.11	1.56	0.02	Slightly Less
Budget	7.00	1.00	3.00	5.00	9.00	1.00	26.00	0.27	Equally Important
							97.92	1.00	Slightly More
									More
									Much More
									Exceptionally More
									9.00

- An Analytical Hierarchy Process (AHP) matrix was used to determine the relative importance of each design objective through pairwise comparisons. Sorting Accuracy & Amount and Budget received the highest normalized weightings (0.27 each), indicating they were considered the most critical factors in system performance and feasibility. Navigation Accuracy followed with a weighting of 0.22, reflecting its importance in ensuring reliable task completion. Simplicity of Design and Efficient Weight Distribution were moderately weighted (0.12 each), while Efficient UI was assigned the lowest priority (0.02), as it does not directly impact core system functionality.

Pugh Matrix:

#	Objective	How to Measure	Weight (higher number is more important)	Gears with Tracks	Aftermarket Rubber Wheels	Self-Manufactured 3D Printed Wheels + Rubber Band Tires	2 Wheel Drive	Steep Slide with Hatch System	Slide With Sorting Wheel	Slide	Gate	Ultrasonic Sensor	Ultrasonic + Infrared Sensor	Scoop pick-up	Claw
1	Simplicity of Design		0.12	-1	2	1	1	1	2	2		-1	0	-2	2
2	Sorting Accuracy & Amount		0.27	0	0	0	2	2	2	0		0	0	0	0
3	Efficient Weight Distribution		0.12	2	1	1	0	-1	0	0		-1	0	0	-1
4	Navigation Accuracy		0.22	-1	1	1	2	0	0	0		0	0	1	-1
5	Efficient UI		0.02	0	0	0	0	0	0	0		0	0	2	1
6	Budget		0.27	-1	-1	2	2	1	2	2		1	2	2	-1
Results															
Weighted Totals				-0.360848139	0.304824277	0.9841136039	1.612636779	0.794644015	1.296718975	0.765696427	0.0289845675	0.5310595254	0.5447612767	0.8796094026	-0.7185838413

- A weighted decision matrix was used to evaluate each concept against the prioritized objectives, applying the normalized weightings to generate final performance scores. The 2 Wheel Drive concept achieved the highest overall score (1.61), followed by Slide with Sorting Wheel (1.30) and Self-Manufactured 3D Printed Wheels with Rubber Band Tires (0.98), indicating strong alignment with the most critical objectives. Concepts such as Claw (-0.72) and Gears with Tracks (-0.36) scored negatively, reflecting trade-offs in complexity, cost, or efficiency relative to project priorities. The results confirm that simpler, lightweight, and cost-efficient mechanisms provide the strongest overall performance within the defined constraints.

Milestone 1 Design Conclusion:

The robot would be driven using two N20 motors provided within the MSE2202 lab kit. The motors would be attached to custom made wheels with rubber brands serving as tires for grip. An ultrasonic sensor and an infrared sensor would be used together to detect the objects and a scoop pick up system would be used to pick up the erasers actuated using one servo provided in the MSE 2202 lab kit. The sorting system will be onboard the robot and consists of a slide into a sorting wheel also driven by a servo also from the MSE 2202 lab kit. The sorting of erasers would be sensed using an infrared sensor to differentiate colours. Finally, the chassis will be designed to accommodate all subsystems of the robot and will be designed in ONSHAPE alongside any subsystem mechanism requiring custom components. Any CAD designs will be manufactured using 3D printed PETG through a 3D printer belonging to one of the group members.